

Quantitative Analysis of Media Bias and Stock Price Dynamics: The 2020 Shock

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Abstract—

I. INTRODUCTION

The news is one of the primary channels through which information about a company reaches the investors who buy and sell its stock, and for that reason the tone of a firm's coverage has long been suspected of moving its share price. When a national outlet casts a firm's quarter as a triumph or its leadership as embattled, the framing travels to thousands of readers who already hold the stock or might soon buy it, and whether that framing leaves a mark on the price or merely repeats what the market has already worked out is a question that reaches well beyond the newsroom. It matters to the investor deciding whether a run of hostile headlines is a signal worth trading on, to the firm weighing how much its public image is worth defending, and to the regulator asking whether the press can tilt the fortunes of the largest companies. The question is also stubbornly hard to answer, because coverage and prices move together for many reasons that have nothing to do with one driving the other, and any convincing account has to pull apart the coverage that carries genuine new information from the coverage that only narrates a move the market has already made.

The sharpest way to draw that line is the efficient market hypothesis [1], which holds that the price of a stock at any given moment already reflects every piece of information publicly available about the firm. Read strictly, this idea leaves no room for the tone of a headline to forecast returns, because whatever the headline reports has been folded into the price by the time it appears in print, and coverage becomes a mirror held up to the market rather than a force acting on it. The hypothesis matters to us not because we expect it to hold to the letter but because it sets the null against which any claim of media influence has to be measured. If the tone of a firm's coverage does help predict its returns, then either the market is slow to absorb public information or the coverage is carrying something the market has not yet seen, and if the tone predicts nothing, then the press is largely reporting the market back to itself. We take no position in advance on which of these

holds, and we do not even assume that influence, where it exists, runs from coverage to prices rather than the other way around, since a firm whose stock has just tumbled tends to draw darker coverage in the days that follow.

The year 2020 offers a rare occasion to put these questions to a genuine test. The onset of the pandemic was a shock that fell on every firm at once, crowded out almost everything else in the news for months, and repriced the whole market within a few violent weeks, so if the relationship between coverage and returns were ever going to bend out of its usual shape, this is the moment it would show. Earlier work has already documented that the coupling between news and markets broke apart around that period [2], which turns the stability of the link from something we can assume into something we have to test directly. We therefore treat the shock as a natural experiment and ask three questions of it. The first is whether the tone of a firm's coverage stepped to a new level once the shock arrived. The second asks the same of prices, whether firm returns settled at a new level after the shock rather than simply riding a market that had been repriced around them. The third, and the one the paper turns on, asks whether the relationship between tone and returns changed its behavior around 2020 even when neither series shifted its average, and in particular whether coverage began to lead returns, or returns to lead coverage, in a way it had not before.

Answering these questions calls for a measure of tone that belongs to a single firm rather than to the market as a whole, so we begin by reading each headline with a target-dependent sentiment model that scores how favorably the headline speaks about the specific firm it names [3], and we collapse those readings into one signed daily figure for each company. For the first two questions we place that figure, and the firm's returns, on a panel regression with fixed effects for every firm and every day, because the plain before-and-after comparison that first suggests itself would confuse a genuine 2020 break with two things that have nothing to do with the pandemic, namely the permanent differences that make some firms more heavily and more warmly covered than others, and the market-wide swings that push every firm in the same direction on

a given day. The fixed effects hold both of these aside and leave the regression looking only at how a firm changed relative to its own past once the day's common movement has been removed. For the third question a single regression is not enough, because the direction of any link is precisely what we want to learn rather than assume, so we turn to a vector autoregression, which treats tone and returns as jointly determined and reads off from the data whether past tone helps predict returns, whether past returns help predict tone, or neither, and we let the timing of any regime change be estimated from each firm's own history rather than pinned to the calendar in advance.

The picture that emerges is quieter than the drama of 2020 might lead one to expect, and that quiet is itself the finding. From an initial pull of 6.28 million headlines we distilled a corpus of 90,579 materially relevant, firm-scored headlines across 26 large United States firms, and on that corpus neither the tone of coverage nor firm returns shows a detectable shift in level after the shock once the movements common to all firms are taken out. The relationship between the two turns out to be real but local. Where coverage and returns do lead one another, the link belongs to individual firms and tends to switch on only after each firm's own structural break, and once the firms are pooled and the broad market is controlled for, no market-wide channel survives in either direction. Our reading is that the 2020 shock did not rewrite the relationship between the financial press and the market so much as leave it as it found it, firm-specific and heterogeneous before the crisis and firm-specific and heterogeneous after, and that a design careful enough to strip out the common shocks earns this conclusion precisely by refusing to manufacture the sweeping aggregate effect the moment seemed to invite.

II. RELATED WORK

A. Media Sentiment and Market Outcomes

A large recent literature establishes that the tone of financial news moves with market outcomes, and the shape of that literature is what places our study. Working directly on the 2020 window, Costola et al. [4] run machine-learning sentiment over COVID-19 news and find a positive relationship between news tone and S&P 500 returns, Huynh et al. [5] tie a constructed feverish-sentiment index to depressed pandemic-era returns, and Bai et al. [6] show that the sentiment-to-return relation itself changes phase by phase across the crisis. A parallel line builds standing aggregate indices: Anastasiou et al. construct a positive-sentiment index from search behavior that cushions G20 return declines [7] and later link vaccine-related sentiment shocks to persistent falls in policy uncertainty and market fear [8]. Others weigh one signal against another, with Verma and Verma [9] finding that economic-news sentiment carries more return-relevant information than the noisier social-media stream, and Eierle et al. [10] showing that an adjusted social-media proxy predicts short-horizon returns once fundamentals are held fixed. A few studies already hint that the effect is uneven across firms: Kim-Hahm et al. [11] report that sentiment moves volume and volatility differently across large

technology names, Sing and Singh [12] find sector- and wave-dependent responses, and Dong et al. [13] document a clear price effect in a high-growth market. What unites this work, and marks its limit for our purposes, is that it reads tone off an aggregate index, a whole market, or a channel; even the studies that glimpse firm variation stop at a handful of names and never attribute tone to the specific firm each headline is about, which is the measurement our study is built on.

B. Measuring Tone at the Level of a Firm

Attributing tone to the right firm is a research problem in its own right, and the tooling has been moving toward it. Domain adaptation came first: FinBERT-style models fine-tuned on financial text read sentiment more faithfully than general-purpose ones [14], [15]. Attribution to a named entity came next. Tang et al. [16] annotate entity spans and their sentiment and show that entity-level resolution beats sentence-level classification, Daudert [17] builds a multi-source entity-level corpus expressly to keep co-occurring names from contaminating one another's scores, and van der Heever et al. [18] pair aspect-based sentiment with a refutation step that filters spurious sentiment-to-return correlations. Firm-level sentiment carries real pricing content in this frame: Mahmoudi et al. [19] tie firm-specific sentiment to corporate-announcement returns, and text-derived sentiment shifts valuation multiples in panel settings [20]. The same literature, though, warns that the measurement is fragile. Prompt wording alone biases language-model sentiment [21], and general-purpose GPT-style models prove highly sensitive to input parameters and stumble on financial entity recognition relative to domain models [22], [23]. We take that warning as a design constraint: rather than let a moving model choice drive the numbers, our pipeline scores each headline for the firm it names with a single target-dependent stance model [3], behind one fixed supervised relevance stage and a transparent signed score, so that every downstream figure traces to one auditable measurement.

C. Dynamic Links and Structural Change

A third strand asks not whether tone and returns are related but in which direction the relation runs, and whether it held through 2020. Glasserman et al. [2] date a structural break in mid-March 2020 after which news and equity markets became markedly less coupled, which turns the stability of the link into something a study has to test rather than assume. On direction the evidence is mixed and time-varying: Nel and van Eyden [24] find causality running mainly from market volatility to sentiment rather than the reverse, while Moutinho et al. [25] detect bidirectional, time-varying causality between sentiment and S&P 500 spreads that is strongest during the COVID crisis, and Cevik et al. [26] trace the same shifting sentiment-to-return dynamic across pandemic waves. Both the volatility and the causality evidence rest on aggregate series. A companion econometric literature supplies the tools and the cautions a break-aware design needs: investor-sentiment series are strongly nonstationary and punctuated by endogenous breaks [27], Granger tests lose validity when parameters drift

unless they are made robust to it [28], and recent panel methods estimate the number, timing, and group structure of breaks directly, including as a way to recover unknown treatment timing such as a pandemic shock [29]–[31], while the COVID cross-section shows breaks and regime shifts in volatility itself [32]. What none of this work does is test the lead–lag relation between firm-level stance and firm-level returns in both directions, on series whose stationarity has been verified and whose break dates are estimated rather than imposed. That combination, firm-level target-dependent stance at scale, level-shift tests for stance and returns under one controlled break, and a break-dated bidirectional dynamic analysis, is the gap this paper fills.

III. DATA COLLECTION

A. Corpus and Coverage

Our object of study is the tone of the news that surrounds large United States firms, so the corpus has to begin as close as possible to the full stream of that coverage. We assemble it from MediaCloud, a service that continuously indexes the output of national news outlets, and we query its United States national collection one firm at a time over the period 2015 to 2025. Each query returns the news items that name the firm, each carrying a publication date, a source outlet, and a headline. We keep the headline as the unit of analysis, because the headline is the one part of a news item written to compress the story’s angle into a single line, and it is therefore where a firm’s coverage wears its stance most plainly. Cast this way the net is deliberately wide: seeded with roughly 250 large-capitalization United States firms, the raw pull returns 6,284,404 headlines, a stream large enough that no single outlet or stretch of time dominates it.

A corpus of this size invites an obvious temptation, which is to make it larger still, and we tested whether a second aggregator, GDELT, could extend the coverage we had already gathered. It could not. The headlines GDELT returned were of visibly poorer quality, often truncated or malformed and heavily duplicated across near-identical records, so folding them into the corpus would have thinned a clean signal with noise rather than deepening it. We keep the study on the MediaCloud stream, whose headlines are consistently well formed and attributed, and treat that stream as the raw material for everything that follows.

One feature of that raw material shapes every later choice. Coverage is spread across firms with extreme inequality: a small number of the most heavily reported names account for the bulk of all headlines, while a long tail of firms surfaces in the news only occasionally. This unevenness is the first thing the data asks us to confront once we begin turning a raw stream of headlines into a panel we can measure.

B. Data Preprocessing

The skew is not a nuisance to be smoothed away; it decides which firms can be studied at all. A daily, firm-level measure of tone is only as steady as the coverage beneath it, and a firm the news names a handful of times a month cannot carry

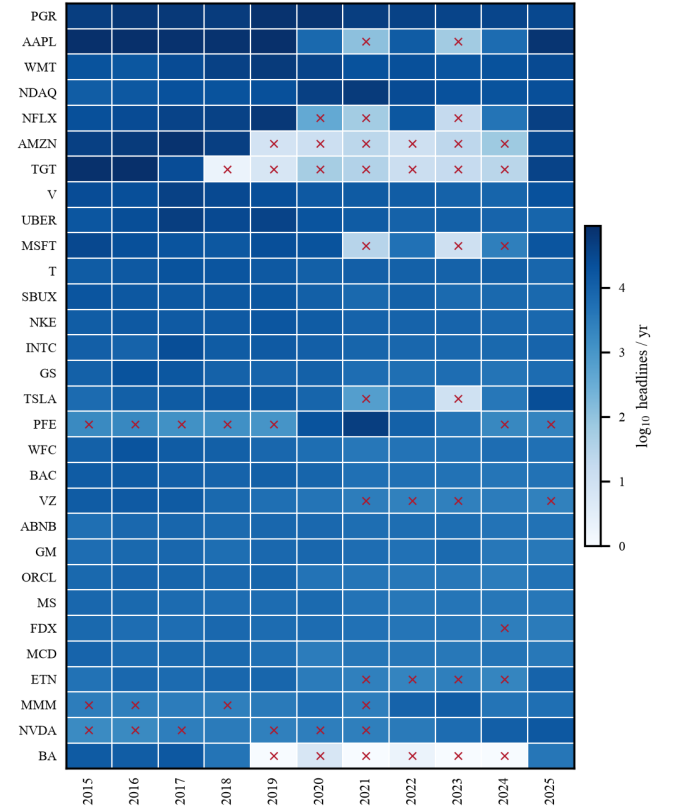


Fig. 1. Yearly headline coverage for each candidate firm, shaded by $\log_{10}$ headlines per year. A cross marks any year in which a firm sits at or below the three-thousand-headline floor; a firm crossed in more than three of the eleven years is set aside as too intermittently covered to support a continuous series.

a daily series that means anything. We therefore keep the firms the press reports on heavily and let the thin ones go. Ranking the universe by the volume of coverage each firm attracts and holding onto the most reported names concentrates the corpus onto its densely covered head, some 4,706,589 headlines, and leaves behind a long tail of firms that never generated enough news to measure.

Volume in a single year is not enough on its own, because coverage also has to persist across the whole decade: a firm that falls quiet for years leaves stretches of time that no series can span. We therefore hold each remaining firm to a floor of more than three thousand headlines in a year, and set aside any firm that slips to or below that floor in more than three of the eleven years from 2015 to 2025, since a firm that runs thin that often cannot anchor a continuous weekly or daily series. Figure 1 maps yearly coverage firm by firm and marks the years that fall below the floor; the firms carried forward are those whose rows stay dark almost throughout, while the ones that flicker pale in year after year are the ones the rule removes.

Two kinds of contamination still sit inside what survives. A great many headlines name a firm only in passing, in a list or an aside, while the story itself is about something else; because

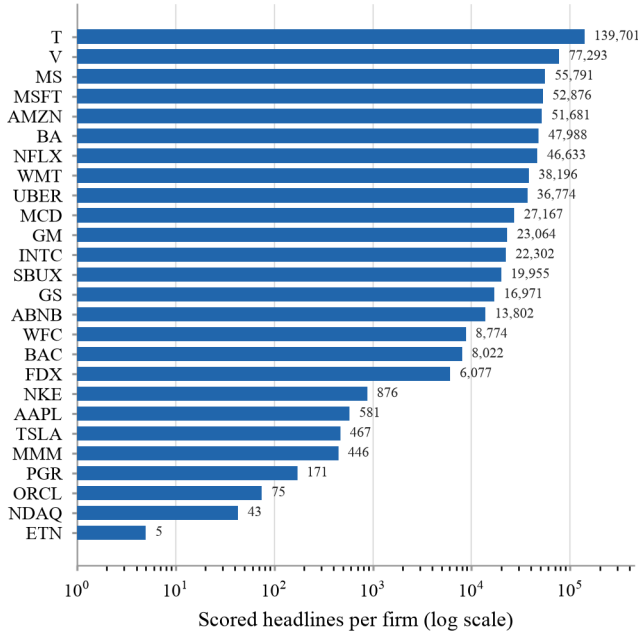


Fig. 2. Scored headlines per firm in the final corpus (26 firms, log scale). Coverage spans more than four orders of magnitude, from AT&T at 139,701 down to Eaton at 5, which the firm fixed effects in the panel regressions absorb.

a headline written about the firm almost always carries the firm’s name or its ticker symbol, we keep only the headlines that do and discard the rest. The same wire report, in turn, is republished word for word across dozens of outlets, so we collapse these near-identical copies to a single record per firm. Table I traces the corpus through these passes, from the full scrape down to what remains: 695,731 scored headlines across 26 firms, each attached to a firm the news followed closely and each naming that firm in its title. Even within this retained set coverage stays strikingly unequal, spanning more than four orders of magnitude from the most reported firm to the least (Figure 2); the estimation later leans on firm-level effects precisely so that these permanent differences in coverage volume cannot masquerade as differences in tone.

TABLE I
FROM THE FULL NEWS SCRAPE TO THE CORPUS THE STUDY RUNS ON.

Stage	Headlines	Firms
Full MediaCloud scrape	6,284,404	250
Most-covered firms retained	4,706,589	30
Name or ticker in title, de-duplicated	695,731	26

IV. METHODOLOGY

A. Relevance Classification

The scored corpus still carries a problem that no amount of name matching can fix: a headline that names a firm is not necessarily about the firm. “Apple” headlines a recipe as easily as an earnings call, banks lend their names to stadiums and

sponsorships, and a firm can drift through a story as scenery rather than subject. If such headlines flow into a firm’s daily tone, the signal we later regress on is diluted by coverage that no investor would read as news about the company. What the study needs is the subset of headlines in which the firm is the primary subject of a material corporate event: earnings, mergers and acquisitions, executive changes, large layoffs, regulatory or legal action, major product launches, or credit and bankruptcy events. “Goldman Sachs beats profit estimates on a trading surge” belongs in that subset; “Ten apple recipes to try this autumn” does not, even though both contain a firm name.

Separating the two at the scale of hundreds of thousands of headlines calls for a supervised classifier, and a classifier calls for labels. We hand-labelled 750 headlines against the written rubric above, resolving borderline cases by review, which yielded 227 relevant and 523 irrelevant examples; a further 187 labelled headlines (145 irrelevant, 42 relevant) were locked away as a held-out test set, touched neither during training nor when choosing the decision threshold. The classifier itself is a DeBERTa-v3-base model fine-tuned for binary classification on the headline text: each headline is tokenized to at most 128 tokens, the model produces a 768-dimensional contextual representation, and a linear head maps that representation to a relevance probability. Training uses the AdamW optimizer at a learning rate of 2×10^{-5} , a batch size of 16, weight decay of 0.01, and early stopping on the validation loss.

Trained on the 750 labels alone, the model reaches a precision of only 0.65 at a recall of 0.71, and the reason is instructive: the labelled set is nearly a third relevant while the true rate in the corpus is closer to one headline in eight, so the model learns to say yes too readily. More manual labels would close that gap slowly and expensively. Instead we widen the training set with 10,025 synthetically generated headlines, built from sector-aware templates so that the events, amounts, and counterparties in each generated headline are plausible for the firm’s industry, and aimed squarely at the categories the base model confuses most: consumer promotions, analyst previews, immaterial lawsuits, and passing mentions on the irrelevant side; earnings, layoffs, and deals phrased in unusual ways on the relevant side. We verify samples of the generated headlines against the rubric and confirm that none duplicates the held-out test set. Training on the combined data uses focal loss, which concentrates learning on the hard boundary cases, and weights each manual label five times a synthetic one, so the synthetic examples stretch the decision boundary without ever outvoting a human judgment.

The finished model justifies the effort. On the held-out test set it reaches an accuracy of 0.94 with a precision of 0.88, a recall of 0.86, and an F1 of 0.87 at the default threshold, and the trade-off between catching relevant headlines and keeping false alarms out can be tuned along the threshold sweep in Figure 3. For full-corpus inference we adopt a stricter operating threshold of 0.80, where precision rises to 0.92 at a recall of 0.81 and accuracy holds at 0.94 (34 true positives, 3 false positives, 142 true negatives, 8 false negatives): when

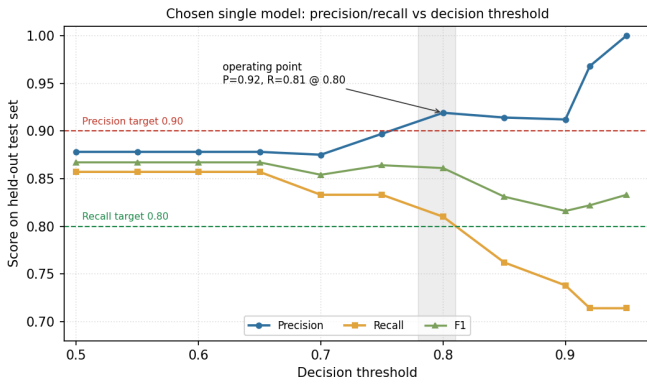


Fig. 3. Precision, recall, and F1 of the final relevance classifier on the held-out test set ($n = 187$) as the decision threshold varies. The operating threshold of 0.80 trades recall for precision ($P = 0.92$, $R = 0.81$), the direction the full-corpus inference favors.

the classifier labels the whole corpus, we would rather lose a genuine story than admit a false one, because every admitted headline feeds directly into a firm's measured tone. Applied to the scored corpus, the classifier retains 90,579 headlines as materially relevant, about 13% of the total, which is consistent with how much of a large firm's news flow reports corporate substance rather than mentions it in passing. These retained headlines are the raw material for every panel that follows.

B. RQ1: Did Media Stance Shift After the Shock?

With a daily, firm-level stance series in hand, the first question the 2020 shock raises is the plainest one: did the tone of a firm's coverage step to a new level once the shock arrived? The tempting way to answer it is to average each firm's stance over the years before 2020 and over the years after, then read off the difference. That comparison would mislead, and seeing why fixes the design for this question and the one that follows.

Two forces move a firm's coverage that have nothing to do with the shock, and a raw before-and-after average folds both of them into what it would label the 2020 effect. Firms differ from one another in ways that persist across the whole sample, so that some are covered more, and more warmly, than others. And on any single day every firm is pushed in the same direction by the news cycle as a whole, so a run of shared bad days, or a change in which firms happen to be in the news, would surface as a break that owes nothing to the pandemic. A panel regression with two sets of fixed effects removes exactly these two confounders. Firm effects tie each firm to its own baseline, so the comparison is between a firm and its own past rather than between one firm and another; daily effects hold constant whatever is common to every firm on a given day, so a market-wide swing in tone cannot be read as a firm-level break. What remains for a post-2020 indicator to explain is the within-firm change net of these common shocks, which is precisely the quantity the question asks about. The estimate is credible under three assumptions the setting makes reasonable: that, absent the shock, the pre- and post-periods would have followed a common trend; that the timing of the break is set

by the pandemic's onset rather than by anything in a firm's own coverage; and that, once the firm and day effects are removed, what is left is uncorrelated with the post indicator. We guard the last of these by reporting HC1-robust standard errors and, because several firms are often covered on the same day, standard errors clustered by day.

The outcome that carries this question is the tone of a firm's coverage on a day. A target-dependent sentiment model reads each retained headline and reports, for the firm the headline names, how likely its tone toward that firm is positive, negative, or neutral; we write these three probabilities as p_a^+ , p_a^- , and p_a^0 for headline a , with $p_a^+ + p_a^- + p_a^0 = 1$. We collapse them into a single signed score, $p_a^+ - p_a^-$, which runs from -1 for wholly unfavorable coverage to $+1$ for wholly favorable coverage and sits near zero when a headline is even-handed or the model is unsure. A firm's stance on a day is the mean of these scores over the headlines it drew that day,

$$\text{bias}_{it} = \frac{1}{N_{it}} \sum_{a \in A_{it}} (p_a^+ - p_a^-), \quad (1)$$

where A_{it} is the set of retained headlines about firm i on day t , $N_{it} = |A_{it}|$ is that day's coverage volume, and $\text{bias}_{it} \in [-1, 1]$ is the average tone toward the firm.

To ask whether this daily stance moved to a new level after the shock, we place it on the post-2020 indicator alongside the firm and daily effects and the day's coverage volume,

$$\text{bias}_{it} = \alpha_i + \delta_t^{\text{day}} + \beta \text{Post}_t + \gamma \text{vol}_{it} + \varepsilon_{it}. \quad (2)$$

Here $\text{Post}_t = \mathbf{1}[t \geq 2020-01-01]$ switches on for every day from January 2020 onward, and its coefficient β is the quantity of interest; α_i are the firm effects and δ_t^{day} the daily effects. The volume control $\text{vol}_{it} = N_{it}$ enters because a daily average tightens mechanically as more headlines fold into it, so holding the count fixed keeps a genuine change in tone from being mistaken for a change in the sheer amount of coverage. We test

$$H_0^{(1)} : \beta = 0 \quad \text{against} \quad H_1^{(1)} : \beta \neq 0,$$

estimating (2) by the within-transformation with HC1-robust standard errors and, since several firms can be covered on the same day, day-clustered standard errors alongside them.

C. RQ2: Did Firm-Level Returns Shift After the Shock?

The second question asks the same of prices. The outcome is firm i 's daily log return, $\text{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$, where P_{it} is the firm's closing price on day t ; we draw the daily closing prices, the S&P 500 index, and the VIX volatility index from `yfinance`. Log returns are additive over time and roughly symmetric, which suits a linear model. Returns carry a large common component: when the market rises, most firms rise with it, so the design must strip that component out before it can see a firm-level break. Monthly effects absorb the broad market regime, and within each month the S&P 500 return and the VIX account for the day-to-day market swings and the level of volatility. The regression is

$$\text{return}_{it} = \alpha_i + \delta_t^{\text{month}} + \beta \text{Post}_t + \gamma_1 \text{sp500_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}, \quad (3)$$

with Post_t as before, α_i the firm effects, δ_t^{month} the monthly effects, and $(\text{sp500_ret}_t, \text{vix}_t)$ the market controls. We again test the shock,

$$H_0^{(2)} : \beta = 0 \quad \text{against} \quad H_1^{(2)} : \beta \neq 0,$$

estimating (3) by the within-transformation with HC1-robust standard errors. Table II gathers every quantity that enters the two regressions, so that the returns model in particular is read as the post indicator and the market controls acting together rather than any one of them alone.

D. RQ3: Did the Stance–Return Relationship Change?

The two level tests leave a question they cannot reach. Learning that neither the tone of coverage nor firm returns stepped to a new level after the shock says nothing about whether the two moved together differently once it arrived; a relationship can change how it behaves while each series holds its average in place. The third question therefore asks not where stance and returns sit but how they lead each other, and whether that lead–lag structure shifted around 2020. We leave the direction of any such link open. Coverage may carry information prices have yet to absorb, so that past stance forecasts returns, or prices may move first and coverage follow, so that returns forecast stance; a design fit for this question has to admit both and let the data choose between them.

A vector autoregression answers to exactly that requirement, because it treats stance and returns as jointly endogenous and reads the direction of predictability off the data rather than off an assumed ordering. For a firm we gather the two series into $\mathbf{y}_t = (\Delta \text{bias}_t, r_t)^\top$ and write the reduced-form VAR of order p as

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^p \mathbf{A}_\ell \mathbf{y}_{t-\ell} + \mathbf{u}_t, \quad (4)$$

with coefficient matrices $\mathbf{A}_\ell \in \mathbb{R}^{2 \times 2}$ and innovations $\mathbf{u}_t$ of covariance Σ . The off-diagonal terms hold the cross-dynamics the question turns on: a nonzero $(2, 1)$ entry means past stance moves current returns, and a nonzero $(1, 2)$ entry means past returns move current stance. We fit the model at a weekly frequency. Daily firm-level stance is too thin for many firms to support a lag-rich dynamic model, whereas aggregating the headlines into calendar weeks restores a dense, near-continuous series for each retained firm while leaving enough weeks for the lags the model needs. We align each weekly stance series to the firm’s weekly log return, bridge the short coverage gaps by imputation, and keep an observed-only version of every series so that no dynamic result can rest on the filled values.

Two properties of the weekly data set the specification. A VAR in levels is trustworthy only when its series are stationary, since a levels regression on integrated data is spurious, so we classify each series by pairing the Augmented Dickey–Fuller test, whose null is a unit root, with the KPSS test, whose null is stationarity. Weekly returns come back stationary for every firm, while the stance index is stationary for some firms and integrated or break-driven for the rest. To hold every firm

to one specification without courting a spurious regression, we take the VAR on the first difference of the stance index, Δbias_t , which is stationary throughout, paired with the weekly return; the handful of firms whose stance and returns test as cointegrated we carry separately as robustness exceptions to the differenced form.

Because the question is about a regime change, we let the data date the change rather than impose the calendar. A Bai–Perron procedure locates the single largest shift in the mean of each weekly series, and that estimated break splits the firm’s sample into a pre- and a post-break segment on which the dynamics can be compared. Dating the break empirically is what makes the comparison honest: some firms break near the pandemic window while others break on news of their own, so a single imposed 2020 cut would misplace the change for many of them. For each firm we set the VAR order by minimizing the Bayesian Information Criterion over $p = 1, \dots, 8$, which penalizes extra parameters heavily and so favors parsimonious, well-identified models, and we certify every fit for stability, with all roots of the companion matrix outside the unit circle, for the absence of residual autocorrelation, and for the heavy tails and volatility clustering typical of financial returns, the last of which is why the inference that follows is made robust to heteroskedasticity and autocorrelation.

On each fitted model we test predictability in both directions through Granger causality, which asks whether the past of one series helps predict the other beyond that other’s own past. Writing $a_\ell^{(2,1)}$ for the coefficient on lag- ℓ stance in the return equation, stance fails to forecast returns when

$$H_0^{(3)} : a_1^{(2,1)} = \dots = a_p^{(2,1)} = 0$$

holds against the alternative that at least one of these coefficients is nonzero, and the mirror-image hypothesis on the $(1, 2)$ coefficients asks whether returns forecast stance. We test each with a joint Wald statistic and Newey–West standard errors that absorb the residual dependence found above, on the full sample and on the pre- and post-break segments in turn.

A firm-by-firm test is underpowered, since each rests on one short series, so the specification the dynamic conclusion leans on pools the firms into a panel VAR. Pooling buys power but revives a hazard: ordinary within-firm demeaning ties the transformed errors to the lagged regressors and bends the estimated dynamics, the Nickell bias. We strip firm heterogeneity with the Arellano–Bover (Helmert) forward transformation instead, which subtracts from each observation the mean of its own future values and so keeps the lagged regressors orthogonal to the transformed error,

$$\tilde{y}_{it} = c_{it} \left(y_{it} - \frac{1}{T_{it}} \sum_{s>t} y_{is} \right), \quad (5)$$

with c_{it} a factor that equalizes the variance of the transformed errors. Into the pooled model we partial the very forces that could otherwise counterfeit a media–market link: the post-2020 indicator, the VIX, the weekly S&P 500 return, and the week’s coverage volume. A Granger relation that survives

TABLE II
VARIABLES IN THE RQ1 (STANCE) AND RQ2 (RETURNS) REGRESSIONS.

Variable	Used in	Meaning
bias_{it}	RQ1 (dep.)	Daily firm stance: the signed mean $p_a^+ - p_a^-$ of the headlines about firm i on day t .
return_{it}	RQ2 (dep.)	Daily log return of firm i , $\ln P_{it} - \ln P_{i,t-1}$.
Post_t	RQ1, RQ2	Shock indicator, $1[t \geq 2020-01-01]$; its coefficient β is the parameter of interest.
vol_{it}	RQ1	Coverage volume: the number of retained headlines about firm i on day t .
sp500_ret_t	RQ2	Daily S&P 500 index return, controlling for broad market movement.
vix_t	RQ2	Daily VIX level, controlling for market volatility.
α_i	RQ1, RQ2	Firm effects, absorbing permanent differences across firms.
δ_t^{day}	RQ1	Daily effects, absorbing shocks common to all firms on a day.
δ_t^{month}	RQ2	Monthly effects, absorbing shocks common to all firms in a month.

Note: “(dep.)” marks the dependent variable; the remaining rows are regressors or fixed effects.

these controls cannot be an echo of market-wide volatility or of how much was written, which is the standard the paper’s central claim is held to.

V. RESULTS

A. RQ1: No Detectable Shift in Media Stance

Estimating (2) on the 27,627 firm-day stance observations, drawn from 26 firms across 3,982 trading days, returns the coefficients in Table III(a). The post-2020 coefficient is $\hat{\beta} = +0.011$ with an HC1 standard error of 0.030 ($p = 0.71$), and clustering by day barely disturbs it ($\text{SE} = 0.030$, $p = 0.72$), so the within-day dependence between firms is not hiding a result. We cannot reject $H_0^{(1)}$: once each firm is set against its own history and the day’s common movement is held fixed, the tone of coverage does not step to a new level after the shock. The volume control behaves exactly as its purpose demands, with a coefficient of -0.0038 ($p < 0.001$), so the days on which a firm is most heavily covered are the days its average tone is pulled back toward neutral, the mechanical averaging effect the control exists to absorb. The within R^2 of 0.002 is what one expects when a single firm’s day-to-day tone is dominated by the noise of individual headlines. A null here is not a failure to find something; it says the shock left no common, firm-level mark on tone once the news cycle’s own daily swings are taken out. Figure 4 puts the signal on view for one firm across the whole window: the tone drifts around a stable, mildly negative mean with no step at the turn of 2020, while the panel beneath it tracks the weekly coverage volume, whose wide swings are the very variation the volume control absorbs.

B. RQ2: No Detectable Shift in Firm-Level Returns

The same estimation on returns, (3) over 20,537 firm-day observations, tells the matching story (Table III(b)). The post-2020 coefficient is all but zero, $\hat{\beta} = -0.0001$ ($\text{SE} = 0.0013$, $p = 0.92$): returns show no level break either. The weekly

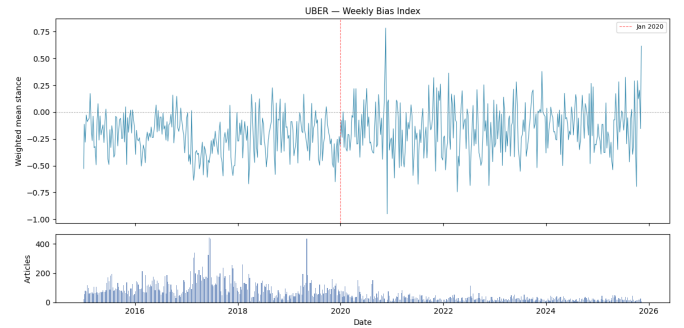


Fig. 4. Stance signal for Uber over 2015–2025. The upper panel plots the coverage-weighted mean stance, which drifts around a stable, mildly negative level with no visible step at January 2020 (dashed line); the lower panel plots weekly headline volume, whose wide swings are the variation the RQ1 volume control is meant to absorb.

return series makes the pattern plain (Figure 5): the March 2020 crash is a violent but brief dislocation, after which volatility settles back toward its earlier range rather than to a new level. What the regression does register is the market itself. The S&P 500 return enters with a coefficient of 1.10 ($p < 0.001$), the familiar market beta near one, and together with the monthly effects it accounts for about a third of the within-firm variation in returns ($R^2 = 0.32$). Volatility adds nothing once the market return is present (VIX, $p = 0.68$). The reading echoes RQ1: firm returns move with the market, as they always have, but the shock itself leaves no separate firm-level step behind. Taken together the two nulls are informative rather than empty; after common shocks are removed, neither the tone of coverage nor firm returns jumped in 2020. A level test is not the only way the shock could have mattered, though: it could have changed how tone and returns move *with one another* over time rather than where each of them sits, and a static regression cannot see that.

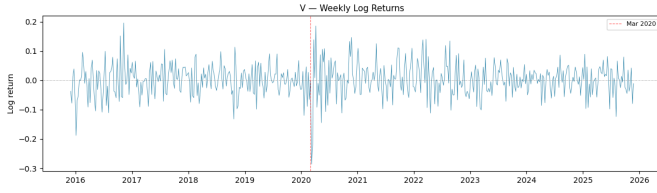


Fig. 5. Visa weekly log returns, 2015–2025. The March 2020 crash (dashed line) is a large but short-lived dislocation, after which volatility returns to its earlier range, consistent with the RQ2 finding of no persistent post-2020 level shift once the market factor is absorbed.

TABLE III

REGRESSION RESULTS FOR RQ1 (STANCE, EQ. 2) AND RQ2 (RETURNS, EQ. 3). BOTH FAIL TO REJECT THE NULL OF NO POST-2020 LEVEL SHIFT ONCE COMMON DAILY (RQ1) AND MONTHLY (RQ2) SHOCKS ARE ABSORBED.

(a) **RQ1 – Daily Stance** (firm + day FE, HC1 SE)

Variable	Coef.	SE	p
Post ($\hat{\beta}$)	+0.01096	0.02964	0.712
Article volume	−0.00381	0.00049	< 0.001

$n = 27,627$; 26 firms; 3,982 days; within $R^2 = 0.0017$
day-clustered $p_{\text{Post}} = 0.717$

(b) **RQ2 – Daily Returns** (firm + month FE, HC1 SE)

Variable	Coef.	SE	p
Post ($\hat{\beta}$)	−0.00013	0.00132	0.921
S&P 500 return	+1.09876	0.01869	< 0.001
VIX	−0.00003	0.00006	0.684

$n = 20,537$; 26 firms; within $R^2 = 0.316$

C. RQ3: A Link That Lives in Individual Firms

Run firm by firm on the full sample, the Granger tests are nearly silent, which is what a regime-dependent relationship would lead us to expect. Only two firms show stance leading returns, Uber decisively ($p = 0.003$) and Boeing weakly ($p = 0.10$), and the reverse channel, returns leading stance, reaches significance for Goldman Sachs alone ($p = 0.004$); for every other firm neither direction clears the five-percent level (Table IV). Taken at face value this reads as a relationship that is barely there.

Splitting each firm at its own estimated break tells a sharper story. Several firms that are quiet over the full sample come alive after their break: Uber’s stance-to-returns channel strengthens from $p = 0.48$ before the break to $p < 0.001$ after it, and Intel turns marginally predictive ($p = 0.074$ post-break). Feedback from returns to coverage surfaces after the break for AT&T ($p = 0.015$), Boeing ($p = 0.010$), and Wells Fargo ($p = 0.030$), while Goldman Sachs and Morgan Stanley show it only before. This pre/post asymmetry is the firm-level counterpart of the structural shift the level tests looked for and did not find: the media–market link is not a fixed feature of the sample but one that switches on after the break for a subset of firms. The estimated break dates are themselves the argument for reading them off the data rather than fixing them by hand

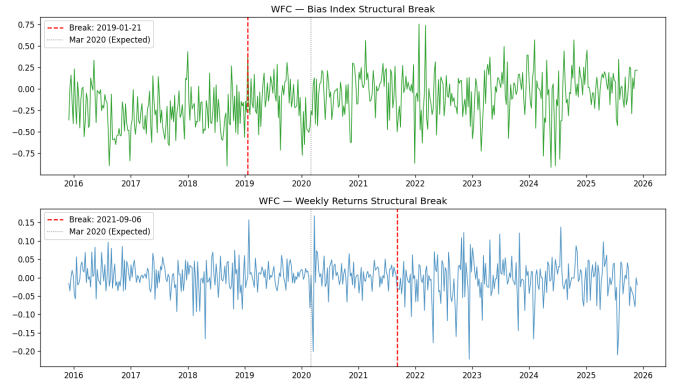


Fig. 6. Data-driven structural breaks for Wells Fargo. The Bai–Perron break in the coverage-stance series (top, January 2019) and in weekly returns (bottom, September 2021) both fall well away from the expected March 2020 date (dotted), which is why the breaks that split the RQ3 subsamples are estimated from each series rather than imposed on the calendar.

TABLE IV

FIRM-LEVEL GRANGER CAUSALITY ON THE FULL SAMPLE (HAC p -VALUES). ONLY FIRMS WITH A SIGNIFICANT DIRECTION ARE SHOWN; THE REMAINING FIRMS CLEAR NEITHER DIRECTION AT THE FIVE-PERCENT LEVEL.

Firm	Lag	Stance→Ret	Ret→Stance
Uber	6	0.003	0.375
Boeing	5	0.097	0.119
Goldman Sachs	3	0.432	0.004

(Figure 6): Wells Fargo’s coverage shifts in early 2019 and its returns only in late 2021, neither near the pandemic, so a break imposed at March 2020 would have split the firm at the wrong moment.

Whether those scattered firm-level signals add up to a market-wide channel is the question the panel VAR settles. With the firms pooled through the Helmert transform, three lags, and the post-2020 indicator, the VIX, the weekly S&P 500 return, and coverage volume all partialled out, neither Granger direction is significant in any subsample: stance-to-returns and returns-to-stance both sit far from conventional levels over the full, pre-break, and post-break windows (Table V). Pooling buys the statistical power the single-firm tests lack, and still nothing aggregate survives once the market-wide controls are in place. The reading is not that the link is absent but that it is local: where predictability exists it belongs to individual firms and their own break dates rather than to the market as a whole. A pooled specification is the design most likely to manufacture a spurious aggregate effect, and here it declines to.

VI. DISCUSSION

A. RQ1: What a Null Stance Shift Means

The absence of a detectable stance shift says something sharper than it first appears. The pandemic was inescapable in the news, yet once each firm is set against its own history and the shocks common to every firm on a day are held aside, the

TABLE V

PANEL VAR GRANGER CAUSALITY, HELMERT-TRANSFORMED AND HAC-ROBUST, WITH POST_t , THE VIX, THE S&P 500 RETURN, AND COVERAGE VOLUME PARTIALED OUT. NO DIRECTION IS SIGNIFICANT IN ANY SUBSAMPLE.

Sample	N	Stance $\rightarrow$ Ret	Ret $\rightarrow$ Stance
Full	7,493	0.565	0.899
Pre-break	4,415	0.301	0.861
Post-break	3,018	0.905	0.937

average direction of its coverage did not lurch. The strongly negative volume coefficient completes the picture: what moved on a firm's heaviest days was how much was written, not how favorably. For an investor or a regulator worried that the crisis bent the press systematically for or against large firms, the daily firm-level evidence offers no support.

The measurement, however, bounds that claim. A signed daily mean is a coarse summary of tone: it can net out opposing headlines within the same day, and it inherits noise from both the relevance filter and the stance model, noise that pulls any true effect toward zero. A single calendar break cannot separate the pandemic from whatever else changed in early 2020. And a headline, however sharply it compresses a story, is not the full article beneath it, so a stance that lived mainly in the body would go unseen.

The natural extensions follow from those limits. A distributional summary of daily tone would keep what the mean erases; carrying the stance model's uncertainty into the regression would let the data quantify the attenuation rather than leave it as a caveat; and scoring the full article text for the most heavily covered firms would measure what the headline restriction costs.

B. RQ2: What a Null Return Shift Means

For returns the null carries a different message. Once the market factor and the broad monthly regime are absorbed, the shock left no firm-specific return signature aligned to the break for the average firm in the panel, which is what efficient pricing of a market-wide event would predict. **flag** The pandemic repriced the market as a whole, and it did so through the channels the market controls already capture rather than through a persistent firm-level wedge that outlives them.

The design is deliberately conservative, and that conservatism is its limitation. By removing the common component it stays silent on the large aggregate repricing that plainly happened, and by testing a shift in the mean it cannot see the moves in volatility and tail risk that were violent in 2020. Loading the broad market onto monthly effects and the index return also risks absorbing part of a genuine firm-level response along with the noise. And the panel treats every firm alike, though the crisis fell on an airline and a grocer in opposite ways.

Each restriction invites a relaxation: modeling conditional volatility directly rather than testing means alone, defining the firm-specific channel against a richer factor benchmark than

the index and the VIX, and trading the single calendar break for event windows drawn around each firm's own news flow.

C. RQ3: A Link That Is Firm-Specific, Not Market-Wide

The dynamic analysis delivers the paper's central interpretive point. Predictability between stance and returns is real, but it lives in individual firms and tends to switch on only after a firm's own structural break; pooled across the panel, with volatility, the market return, and coverage volume held aside, no market-wide lead-lag channel survives in either direction. A pooled test is the design most able to conjure an aggregate effect, and it declines to, so the honest reading is heterogeneity rather than absence.

The caveats are those of any reduced-form dynamic analysis. Weekly aggregation, forced by how sparse daily firm-level stance is, may alias away a faster channel through which headlines are priced within days or hours. The bivariate design carries only stance and returns, so drivers that move both, among them earnings surprises, analyst actions, and sector shocks, sit outside it, and the panel controls narrow that gap without closing it. A single estimated break describes a firm with several regime changes only in part, and the subsample split treats a break date that was itself estimated as though it were known. Heavy-tailed, volatility-clustered residuals leave the inference resting on the HAC correction and large-sample approximation rather than on exact distributions. And Granger causality is predictive, not structural: a rejection says past stance forecasts returns, not that stance moves prices.

The most promising directions answer those caveats in turn: pushing to daily or event-time frequency for the densely covered firms, allowing several breaks or smoothly switching regimes in place of one, folding the omitted drivers into the model as exogenous regressors, and pairing the reduced-form evidence with an identification strategy that can speak to mechanism rather than to prediction alone.

VII. CONCLUSION

We set out to learn whether the 2020 shock changed what the news says about large United States firms, what their stocks do, and how the two move together. From 6.28 million headlines we distilled a corpus of 90,579 materially relevant, target-scored headlines across 26 S&P 500 firms, and put the shock to three tests: pre/post panel regressions for the level of stance and of returns, and a break-dated weekly dynamic analysis for the link between them. The levels did not move: neither media stance nor firm-specific returns shows a significant aggregate shift once common shocks are absorbed. The link between stance and returns turns out to be real but local, surfacing in individual firms after their own structural breaks while no market-wide channel survives the pooled, controlled test. The 2020 shock, on this evidence, did not rewrite the relationship between the financial press and the market; it left a relationship that was firm-specific before the crisis just as firm-specific after it.

REFERENCES

- [1] E. F. Fama, "Efficient capital markets: A review of theory and empirical work," *The Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.
- [2] P. Glasserman, H. Mamaysky, and M. Shen, "News and markets in the time of COVID-19," *Journal of Financial and Quantitative Analysis*, 2024.
- [3] F. Hamborg and K. Donnay, "NewsMTSC: A dataset for (multi-)target-dependent sentiment classification in news articles," in *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2021, pp. 1663–1675.
- [4] M. Costola, O. Hinz, M. Nofer, and L. Pelizzon, "Machine learning sentiment analysis, COVID-19 news and stock market reactions," *Research in International Business and Finance*, vol. 64, 2023.
- [5] T. L. D. Huynh, M. Foglia, M. A. Nasir, and E. Angelini, "Feverish sentiment and global equity markets during the COVID-19 pandemic," *Journal of Economic Behavior & Organization*, vol. 188, pp. 1088–1108, 2021.
- [6] C. Bai, Y. Duan, X. Fan, and S. Tang, "Financial market sentiment and stock return during the COVID-19 pandemic," *Finance Research Letters*, vol. 54, 2023.
- [7] D. Anastasiou, A. Ballis, and K. Drakos, "Constructing a positive sentiment index for COVID-19: Evidence from G20 stock markets," *International Review of Financial Analysis*, vol. 81, 2022.
- [8] D. Anastasiou, A. Ballis, C. Kallandranis, and I. Vlassas, "Positive COVID-19 related sentiment, economic uncertainty & risk management implications," *Journal of Banking Regulation*, 2026.
- [9] R. Verma and P. Verma, "Economic news, social media sentiments, and stock returns: Which is a bigger driver?" *Journal of Risk and Financial Management*, vol. 18, no. 1, 2025.
- [10] B. Eierle, S. Klamer, and M. Muck, "Does it really pay off for investors to consider information from social media?" *International Review of Financial Analysis*, vol. 81, 2022.
- [11] H. Kim-Hahm, A. S. Abou-Zaid, and A. Mohd, "News vs. social media: Sentiment impact on stock performance of big tech companies," *Journal of Risk and Financial Management*, vol. 18, no. 12, 2025.
- [12] N. B. Sing and R. G. Singh, "Investor attention and reaction in COVID-19 crisis: sentiment analysis in the indian stock market," *Managerial Finance*, vol. 49, no. 4, 2023.
- [13] X. Dong, S. Xu, J. Liu, and F.-S. Tsai, "Does media sentiment affect stock prices? evidence from china's STAR market," *Frontiers in Psychology*, vol. 13, 2022.
- [14] D. Araci, "FinBERT: Financial sentiment analysis with pre-trained language models," arXiv preprint arXiv:1908.10063, 2019.
- [15] Z. Liu, D. Huang, K. Huang, Z. Li, and J. Zhao, "FinBERT: A pre-trained financial language representation model for financial text mining," in *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.
- [16] Y. Tang, Y. Yang, A. Huang, A. Tam, and J. Tang, "FinEntity: Entity-level sentiment classification for financial texts," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- [17] T. Daudert, "A multi-source entity-level sentiment corpus for the financial domain: the FinLin corpus," *Language Resources and Evaluation*, vol. 55, 2021.
- [18] W. van der Heever, K. Ong, R. Satapathy, and E. Cambria, "Beyond correlation: Refutation-validated aspect-based sentiment analysis for explainable energy market returns," arXiv preprint arXiv:2603.21473, 2026.
- [19] N. Mahmoudi, P. Docherty, and A. Melia, "Firm-level investor sentiment and corporate announcement returns," *Journal of Banking & Finance*, vol. 144, 2022.
- [20] Foreign Trade University, "Analyzing and forecasting P/E ratios with investor sentiment in 2025," Foreign Trade University, Working Paper, 2025.
- [21] V. Gandhi and S. Gandhi, "Prompt sentiment: The catalyst for LLM change," arXiv preprint arXiv:2503.13510, 2025.
- [22] London Stock Exchange Group, "Using GPT-4 with prompt engineering for financial industry tasks," LSEG, White Paper, 2023.
- [23] X. Li, S. Chan, X. Zhu, Y. Pei, Z. Ma, X. Liu, and S. Shah, "Are ChatGPT and GPT-4 general-purpose solvers for financial text analytics? a study on several typical tasks," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2023.
- [24] P. Nel and R. van Eyden, "From news to noise: Does media sentiment drive stock market volatility?" University of Pretoria, Working Paper 202605, 2026.
- [25] V. Moutinho, R. H. C. Domingues, G. Fantini, and M. Moraes, "Examining time-varying causality: investor sentiment and asset spreads across COVID and Ukraine war periods," *Applied Economics Letters*, 2025.
- [26] E. Cevik, B. Kirci Altinkeski, E. I. Cevik, and S. Dibooglu, "Investor sentiments and stock markets during the COVID-19 pandemic," *Financial Innovation*, vol. 8, 2022.
- [27] D. Ballinari and S. Behrendt, "Structural breaks in online investor sentiment: A note on the nonstationarity of financial chatter," *Finance Research Letters*, vol. 35, 2020.
- [28] B. Rossi and Y. Wang, "Vector autoregressive-based Granger causality test in the presence of instabilities," *The Stata Journal*, vol. 19, no. 4, 2019.
- [29] J. Ditzen, Y. Karavias, and J. Westerlund, "Testing and estimating structural breaks in time series and panel data in Stata," *The Stata Journal*, 2025.
- [30] F. Pretis and M. Schwarz, "Discovering what mattered: Answering reverse causal questions by detecting unknown treatment assignment and timing as breaks in panel models," SSRN Working Paper, 2026.
- [31] R. L. Lumsdaine, R. Okui, and W. Wang, "Time-varying group heterogeneity in panel data models," *Journal of Econometrics*, vol. 232, no. 1, 2023.
- [32] A. M. B. Oliveira, A. Mandal, and G. J. Power, "Impact of COVID-19 on stock indices volatility: Long-memory persistence, structural breaks, or both?" *Annals of Data Science*, vol. 11, 2024.